

Human-Centered and Sustainable Artificial Intelligence in Industry 5.0: Challenges and Perspectives

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Abstract: The aim of this position paper is to identify a specific focus and the major challenges related to the human-centered artificial intelligence (HCAI) approach in the field of Industry 5.0 and the circular economy. A first step towards the opening of a line of research is necessary to aggregate multidisciplinary and interdisciplinary skills to promote and take into consideration the different aspects related to this topic, from the more technical and engineering aspects to the social ones and the repercussions in terms of sustainability. The proposal and vision of this preliminary work is to identify and discuss a suitable field for such interaction. This field has been identified, specifically, within additive manufacturing (AM) in the context of Industry 5.0. Additive manufacturing (AM), is a disruptive opportunity for more sustainable production systems that can be better optimized with AI, becoming an ideal platform for interconnection between different levels of application and integration of HCAI concepts, and at the same time able to prove them. In this context, two prospective areas with a high application impact of HCAI are those of AM-oriented supply chain and product customization in the AM field, enabled by a plethora of recently emerging technologies such as the internet of things, cloud and edge computing, and next-generation networks (5G). The paper concludes with the challenges HCAI poses to public policymakers, who face significant policy challenges in regulating artificial intelligence, and addressing the socioeconomic and technological impacts. Decision-makers are required to address these challenges by adopting some tentative policy recommendations.

Keywords: human-centered artificial intelligence; Industry 5.0; enabling technologies; additive manufacturing; customization; regulation; public policy



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1. Introduction

Artificial intelligence (AI) is a field of study that focuses on creating systems or machines capable of performing tasks that require human intelligence. These tasks include machine learning, problem-solving, pattern recognition, natural language processing, and more. In general, AI aims to develop systems that can simulate some of the human capabilities of learning and reasoning, such as visual perception, voice recognition, and linguistic translation, to perform specific tasks autonomously or semi-autonomously. John McCarthy specifically described AI as the scientific and technological competence for the development of intelligent computer programmers [1].

Machine learning (ML) and deep learning (DL) are two of the most used methods of artificial intelligence [2]. These models rely on data and are used to develop predictive models by individuals, companies, and governmental organizations. Currently, methods of automated learning are being developed capable of handling the complexity and unpredictability of information in various industrial sectors such as food, biomedical, and aerospace [3]. By combining the principles of ML and DL with advanced optimization techniques into industrial processes, practitioners are empowered to navigate modern challenges with heightened precision and efficacy [4–8].

At the industrial level, this has led to the development of advanced approaches to cognitive computing and deep learning for automated applications such as visual inspection, fault detection, and maintenance in manufacturing systems. Deep learning approaches are actively used in production systems, from supply chain to manufacturing programs, ensuring the highest quality and safety in multiple sectors [9], including public policy and public administration.

AI helps to monitor and manage production processes comprehensively and not only to replace humans in risky operations but also to support them, placing them at the center of the process through a specific approach called human-centered. Numerous examples include the adoption of human-centered approaches by major companies such as Apple, Google, and Microsoft during the development of artificial intelligence software [10]. Mercedes-Benz has replaced standard robots with AI-based collaborative robots (cobots), enabling the production of customized cars more efficiently [11]. IBM Watson has proposed a system that could recommend cancer treatments in line with the doctor's recommendations most of the time [12]. Furthermore, AI and sensors offer real-time technological advancements that support the shipping industry, enhancing safety, reducing costs, thereby boosting productivity in international trade and embracing a more intelligent and sustainable future [13,14].

This underlines the importance of a multidisciplinary approach that allows different sectors and roles to communicate and highlight positive aspects and diverse perspectives of AI use. This article explores the key challenges associated with the human-centered artificial intelligence (HCAI) approach in Industry 5.0 and the circular economy, as well as the role of decision-makers. The methodological approach will be integrative and comprehensive, employing qualitative methods to examine the impact and applications of HCAI in these fields. This involves conducting an extensive literature review, critically analyzing major challenges with a focus on smart manufacturing and the role of policymakers, and identifying future research trends and needs. As a guiding framework for our study, we set the following research questions (RQs):

1. What does the human-centered AI approach mean in the context of the industrial sector, in general, and in the manufacturing sector in particular?
2. What is the associated literature of the human-centered AI approach when realizing the vision of Industry 5.0 and the main challenge to address?
3. What role do decision-makers play in the successful adoption and implementation of HCAI in Industry 5.0 and the circular economy?
4. How can interdisciplinary collaboration enhance the effectiveness of HCAI applications in industry?
5. What are the future directions for research and development in the field of HCAI?

The remainder of the paper unfolds as follows: the theoretical framework is outlined in Section 2, while Section 3 details the HCAI and Industry 5.0 and provides empirical evidence on the perspectives of HCAI for additive manufacturing (AM). Section 4 discusses the challenges for Government 5.0 in the light of the HCAI approach. Section 5 proposes some suggestions for further research.

2. Background

2.1. Human-Centered Artificial Intelligence

“Human-centered AI” or “human-centered artificial intelligence” refers to a model of AI development and implementation that prioritizes human needs, values, and perspectives. Essentially, it is an approach within the field of AI that focuses on engaging people with active inclusion of users, stakeholders, and industry experts throughout the entire AI development cycle, from design and development phases to practical implementation. The aim is to understand and integrate human perspectives into the decision-making and system creation processes of AI. Facilitating human-machine interaction aims to create AI systems that are intuitive, understandable, and easily usable by humans, reducing the gap and improving interaction between people and technology. The goal of human-centered AI

is to improve people's lives; for example, through solutions that enhance efficiency, accessibility to services, safety, health, and overall quality of life. A schematic representation of the principal aspects involved in human-centered AI is reported in Figure 1.

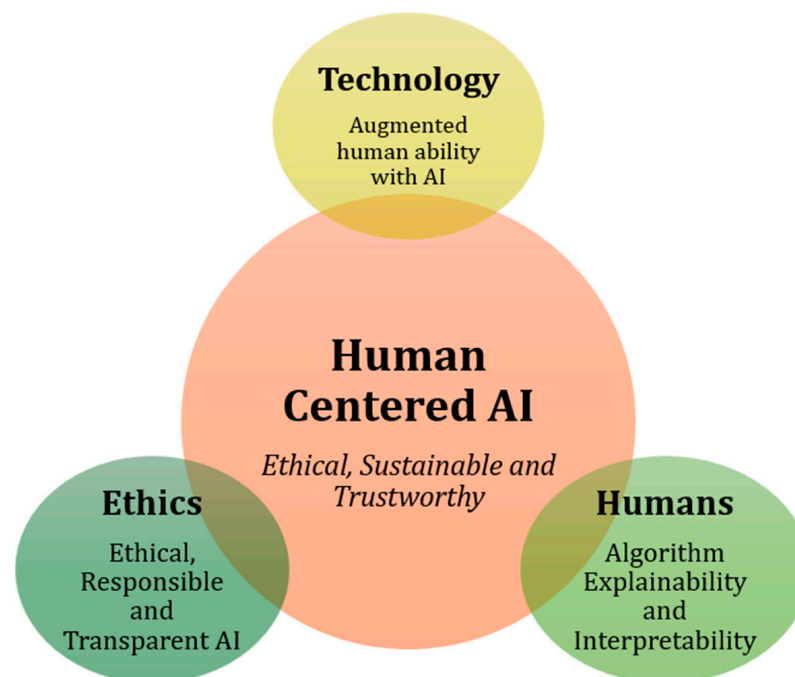


Figure 1. Human-centered AI involves different aspects such as technology, ethics and human factors (based on [15]).

Ethics and transparency are also crucial themes in human-centered AI: human-centered AI places particular emphasis on ethics, responsibility, and transparency in the use of AI systems. The goal is to ensure that algorithms are developed and used responsibly, respecting human values, avoiding discrimination, and making AI decision-making processes transparent. In summary, human-centered AI places humans, their values, and their needs at the center, promoting the adoption of AI technologies in an ethical, responsible, and beneficial manner for humanity.

These considerations stem from the observation that AI has had many failures, such as Facebook's chatbot, which could only correctly respond to 30% of its Messenger services, and Microsoft's AI chatbot, which learned racist insults within a day based on reading Twitter feeds [16,17]. Although these systems may have met all functional requirements and technical goals, the outcome did not reflect the human-centered needs of the users. For example, a technical goal to build a fast facial recognition system could be easily achieved [18]. However, the resulting system could still discriminate based on the color or race of users [19]. These human-centered aspects should be addressed alongside technical goals. In the context of AI, human-centered approaches include goals such as providing a better user experience, greater clarity and usability of information, fairness, trust, bias reduction, and building responsible AI [20–22]. However, today's AI software lacks these human-centered aspects, and appropriate AI solutions need to be researched before including them in software systems.

In the industrial and manufacturing sector, artificial intelligence (AI) offers tremendous potential for development in terms of efficiency, sustainability, and competitiveness. A human-centered AI approach is essential to ensure that engineering technologies and systems are safe, ethical, and beneficial for the people involved. There is a need for a significant shift in perspective within the industrial-manufacturing sector. Initially, the adoption of AI focused on engineering, computer science, and statistics, where the emphasis was primarily on data management and analysis to improve production processes.

Manufacturing companies gather vast amounts of unstructured data from various sources such as sensors in machinery, production lines, production execution systems, enterprise resource planning systems, and systems outside production (customer feedback, supply chain). Analyzing these data creates a competitive advantage and generates new products and services. The ability to properly analyze and structure the collected data and derive value from it through AI and machine learning/deep learning techniques is therefore a challenge for the global industry and the global economy. Countries worldwide are implementing strategies and initiatives to keep pace with change through rapid innovation development and digitalization of production [23].

However, our recent literature review has found that most AI software systems lack human-centered approaches during the writing and modeling of engineering requirements. For example, it has highlighted that many initiatives primarily focus on some aspects to include ethics, trust, and clarity, but lack the human-centered aspects adopted by the industry, or how they should be addressed. A focus on centrality leads to the development and processing of AI-based system requirements in a new and in-depth manner to address human-centeredness. There are indeed engineering requirements of AI that raise a number of challenging issues with which it is difficult to interface, such as placing full trust in AI solutions, specifying various requirements because they are too vague, limitations of existing techniques to manage AI requirements, as well as the emergence of new types of requirements, such as data, clarity, transparency, compliance, and difficulties in understanding the feasibility of AI models.

In the manufacturing industry, AI should not only focus on process optimization but also on creating safe and ethical work environments, considering issues of privacy and data security, and actively involving operators and workers in AI-related decisions. Therefore, in addition to engineering, computer science, and statistical aspects, the issue needs to be addressed from legal, social, and economic perspectives with appropriate policies and legislation to support a new vision.

Another field of application of AI is the public sector, which represents the largest market for AI solutions in developing countries: AI governance becomes important especially now that open government and digitalization of public services are taking root [24]. AI in the public sector is applied, for example, to decisions regarding public procurement, whose spending represents up to 50% of GDP in most cases in developing countries. Therefore, public procurement is essentially a critical bridge for the adoption of AI technologies by the public sector [25,26]. However, governments of developing countries have weak regulatory and governance mechanisms to incentivize solution developers and AI users to adopt human-centered AI innovations that protect users from abuses, social divisions, and government suppression.

2.2. HCAI and Industry 5.0

The European Commission proposes a vision of Industry 5.0, where well-being and technological progress are considered jointly from a sustainability perspective with particular attention to human centrality [27]. The Industry 5.0 revolution requires the intervention of more sophisticated systems such as network sensor data interoperability, cobots, and other intelligent systems [28]. In this context, it is necessary to consolidate knowledge about human-technology interaction to bridge the gap between production and societal needs. AI is not a cold replacement for labor but an opportunity for transformation and growth [29]. In this direction, human-centered AI (HCAI) advocates for a human-centered approach in realizing the vision of Industry 5.0 to create AI systems that enhance human capabilities rather than replacing them and ensure greater inclusivity in line with sustainable development [30,31]. While several works in the literature show examples of AI systems with substantial impacts on human activities, there is unanimous agreement on the need to address all social implications of this technology and how AI can serve social goals with the worker's well-being placed at the center of the production process [32–34]. With a focus on human-machine interaction, Industry 5.0 interconnects human intelligence

with the precision and efficiency of machines using artificial intelligence in industrial production. Industry 5.0 is developed to overcome the challenges faced by Industry 4.0 by promoting human centrality and meeting societal needs. Europe has taken a leading role in the green and digital transitions, which implies that workers, regions, and societies are facing extremely rapid transformations. While these transformations create opportunities for inclusive technological and social development, they also pose the risk of increasing inequalities. The request from the European Commission is to focus efforts on promoting human centrality in AI as an opportunity for transformation and growth to contribute to achieving the goals of the Digital Decade program, which aims to lead Europe's digital transformation by 2030 [35].

Human-centered AI (HCAI) in Industry 5.0 is a new approach aimed at using artificial intelligence systems with a focus on the importance of collaboration and interaction between intelligent machines and human operators. This approach learns from human inputs and is based on collaborative principles that, in addition to technology, draw on knowledge from human sciences (from ethics to behavioral disciplines) so that collaboration and synergy between humans and AI can develop in an environment of growing trust. Examples include machines and processes designed considering the needs, skills, and preferences of human operators and learning from their behaviors: machines assist and support human operators, improving productivity and workplace safety; machines continually learn from human interactions, adapting to the evolving needs of operators and the work environment [36].

In its policy brief on Industry 5.0, the EU not only defines what Industry 5.0 is but also outlines the policies to be implemented to support its development. In the same document, the EU establishes the three axes around which Industry 5.0 develops, namely human-centricity, sustainability, and resilience [35]. In particular, the EU emphasizes the importance of prioritizing research and innovation as engines for the transition to Industry 5.0. Now more than ever, it believes it is important to invest to overcome the economic challenges posed by the coronavirus crisis and to establish a "new normal" with a more competitive, sustainable, and greener European industry. The role of the EU is to lead this new wave of innovation, as reflected in the Next Generation EU and Horizon Europe programs.

This industrial transformation requires a new type of entrepreneur. The entrepreneur of the future will be someone sensitive to innovation, capable of identifying new opportunities for sustainable business and environmental care, and actively participating within an ecosystem of industries and public and private entities to be part of a cooperation model aimed at implementing good practices of the circular economy.

Industry 5.0 aims to revitalize the presence of human labor in factories, where humans and machines would work together to increase process efficiency by fully leveraging human intellectual abilities and creativity through their integration with current intelligent systems. Industry 5.0 includes the interoperability of network sensor data with new added functionalities. Some of these include smart additive manufacturing, predictive maintenance, hyper-customization in industry, cyber-physical cognitive systems, and the introduction of collaborative robots. In smart factories inspired by the modern manufacturing industry, the use of AI facilitates timely decision-making based on both real-time and historical data, with minimal human involvement [36].

To address these challenges, it is necessary for AI to be properly understood, accepted, and integrated, also resolving issues related to data security. If these challenges are effectively overcome, they will help organizations grow Industry 5.0. This will also increase confidence in automation and help workers perform their jobs alongside robots. Better automation that considers human behavior and needs will help improve human productivity and contribute to providing meaningful work to humans themselves. Therefore, Industry 5.0 offers an opportunity to conduct research, particularly in data security and integration, considered the most significant challenge when things are integrated with the internet, often referred to as the industrial internet of things (IIoT). In addition to personal-

ized human-machine interaction, new production and industrial processes enhanced by interaction with AI and IoT are certainly represented by additive manufacturing. So-called additive manufacturing (AM) is a smart technique that falls within the fundamentals of both the fourth and fifth industrial revolutions. Artificial intelligence and machine learning are becoming increasingly integral to the growth and applicability of AM. The potential of this manufacturing technology is currently at the forefront of a scientific and technological revolution. The manufacturing logic is revolutionized, as well as the human role and human-machine interaction. This extends not only to the actual production phase but also to pre- and post-production phases, with a complex revolution of the AM supply chain, as well as the development of new materials, machine improvements, part design, and workflow optimization. With increased human-machine interaction, future work can also be done to develop intelligent control by refining current pattern recognition algorithms.

Similarly, Toth et al. have summarized the potential advantages of the human-centered Industry 5.0 collaboration architecture: improved collaboration, enhanced decision-making, user-oriented design, flexibility, and continuous innovation. They also identified some challenges, such as technological complexity, the need for training and adaptation, financial implications, and the risk of technological redundancy [37].

The core values of Industry 5.0, including human centrality, sustainability, and resilience, have stimulated formal and scientific discussions on the topic. Human-centered smart manufacturing (HSM) fully exploits human flexibility, machine precision, and next-generation computing technologies to build an ultra-intelligent, sustainable, and resilient production system. Research on HSM is relatively lacking but is rapidly gaining ground. For example, Zhang et al. summarized the limitations, barriers, and challenges of key enabling technologies and the application of HSM based on two aspects: they identified three promising Industry 4.0 technologies, such as blockchain and IIoT, human-cybernetic fusion based on DT and AR, and data- and knowledge-assisted intelligent decision support, which can continue to support the construction and operation of HSM. On the other hand, challenges remain related to data security in the IIoT network, the efficiency of collaboration in HCPS (human-centric production service), and model uncertainty in the decision-making process. Zhang et al. also report new applications in HSM and how HSM will reshape the product's lifecycle from the design and production to maintenance phases of manufacturing. By placing the needs, perspectives, and involvement of human operators at the center of decisions and processes, each stage can be optimized as described in the following cases (Figure 2) [38].

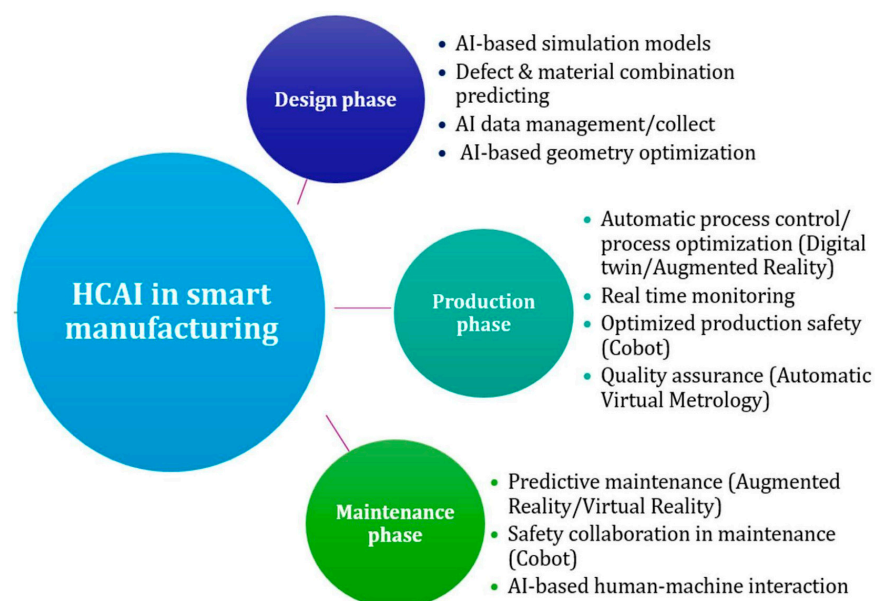


Figure 2. Human-centered AI in smart manufacturing (based on [38]).

1. **Design Phase:** Enhanced decision-making capacity and a robust knowledge base, built on real-time and historical data, can significantly benefit the design phase [39,40]. In the system requirement gathering, the goal is to iteratively translate customer needs into practical and applicable solutions. For example, when designing a production process for manufacturing automotive components using a data-driven HCAI approach, requirement gathering can benefit from predictive analytics made about customer needs and formalization of requirements using data mining techniques [41]. Additionally, design activities can be facilitated by advanced AI techniques and data management methods to analyze collected data, identify patterns, anomalies, and optimization opportunities in the production process, thereby detecting inefficiencies in operator movements or suggesting production parameter changes to enhance quality. Furthermore, AI-based simulation models can be utilized to assess proposed changes to the production process before their actual implementation, predicting any potential negative impacts on performance or worker safety. Additionally, AI algorithms and historical data can optimize raw material combinations and the geometry of AM products, accounting for various design variables and their intricate interactions [42,43].
2. **Production Phase:** AI plays a crucial role in optimizing human-machine interactions within industrial production settings, enhancing the production phase. Data and knowledge-driven production systems, such as blockchain IIoT systems, deep learning, knowledge reasoning, and visualization technologies (dashboards), enable real-time monitoring of production flow and data collection on performance, cycle times, and product quality. Deep learning techniques and cognitive reasoning analyze this data to identify significant patterns and trends, facilitating rapid and informed decision-making. For instance, data analysis may reveal that a specific configuration of machines and optimized cycle times can significantly enhance production efficiency while ensuring better working conditions or reducing worker burdens [44,45]. Moreover, workers can benefit from a more detailed and informed perception of the work environment and production context through AI technologies like digital twins (DT) and augmented reality (AR), improving their decision-making capacity and ability to adapt to evolving conditions. For example, workers can wear AR devices providing real-time information on production process details, such as assembly instructions overlaid on the piece they are assembling, technical specifications, and safety alerts. Additionally, collaborative robots equipped with advanced sensors and machine learning capabilities can detect workers' emotional, physical, and mental states and cognitive biases to make adaptive adjustments, maximizing worker well-being and fundamental rights protection. In terms of workplace safety, this can have significant implications; for example, if a worker is found to be subject to stress or deficits related to compensatory recovery, the robot can adapt its work pace to avoid overloads or risks of accidents [46].
3. **Maintenance Phase:** AI can significantly enhance the maintenance phase by fostering collaboration between human operators and intelligent machines (robots) for effective machinery upkeep. In intelligent planning, maintenance technicians, machine operators, and engineers collaborate, with AI aiding decision-making and optimizing maintenance activities according to human needs. Intelligent machines can perform complex tasks or work in hazardous environments, while technicians handle more flexible tasks. Additionally, augmented reality (AR) systems can be developed to provide detailed instructions and virtual visualizations to guide operators through complex maintenance procedures directly in the field, making it easier to perform such maintenance more accurately and, above all, safely. AI can power AR systems with updated and relevant information, improving the efficiency and accuracy of maintenance activities performed by operators [47,48]. Advancements in natural language processing and gesture recognition enhance machines' ability to interpret human actions, facilitating the transfer of human skills to manufacturing [49]. Above

all, to make human-machine collaboration more effective and safer, it is necessary to have systems that adapt to the needs, preferences, and individual requirements of operators, such as heart rate, operation purpose, and level of experience [50]. In this regard, miniaturization and reduction of costs have increasingly enabled the adoption of wearable sensors in the industrial context to trace workers' conditions and well-being; e.g., eye-trackers can be used to estimate workers' attention [51].

AI technology is crucial in promoting manufacturing innovation, as smart manufacturing processes or industrial upgrading, especially in the development of new technologies and materials. AI technologies, once applied to process manufacturing, allow optimizing the production cycle by automating tasks, predicting outcomes, and improving efficiency. Specifically, they are used from optimizing the supply chain to predicting equipment failures, reducing waste, improving production times and quality. The influence of AI on manufacturing technologies is not limited to the production process, but also to process and design new raw materials. Papadimitriou et al. studied the recent developments in the application of AI to materials design and discovery, including ML, deep learning (DL), and other AI techniques [52]. Moreover, other AI techniques such as evolutionary algorithms (EA) have been used in materials design to optimize the synthesis of materials [53].

Specifically, what this document focuses on is how artificial intelligence is significantly influencing additive manufacturing (AM), which is already a smart technology in itself, and how human-centricity plays an important role in it. The role of AI in technologies and materials is undisputed and the subject of numerous studies

In the field of additive manufacturing, there has been a growing demand for products and devices oriented towards human well-being, such as biomedical products for orthopedics, sports, and dentistry, due to the significant level of customization required by the manufacturing process and at the same time high-quality and cost competitiveness. The use of AI can help overcome challenges for customizing the additive manufacturing process by integrating AI/ML systems in the design, production, and process and product evaluation phases [48,54]. Enabling AI in the product design phase offers a promising solution to consider a variety of design variables and their complex interactions in design and thus achieve desired performance in production. Furthermore, with the integration of AI in the AM manufacturing and evaluation phases, the manufacturing process of human-centric customized products can be optimized, and quality performance can be fully and efficiently evaluated. There are still many open challenges in including AI in additive manufacturing processes, particularly in quality control, allowing for more efficient, precise, and reliable product production [50], especially in the following areas. (a) Quality prediction: using machine learning algorithms, critical quality characteristics such as defects, geometric deviations, and process conditions can be predicted before they actually occur during production. (b) Prescriptive compensation and correction: AI can be used to automatically compensate for any defects or deviations during manufacturing, ensuring that the final product meets the required quality standards [43,55].

3. Development and Perspectives of HCAI for Sustainable Manufacturing Focusing on Additive Manufacturing

The most recent literature review states that additive manufacturing is playing and will play a role in the transition to greater sustainability in manufacturing, as the design and opportunities it offers foster more sustainable production and consumption [56]. This allows substantial economic and environmental benefits, and at the same time promotes sustainable consumption yet collaborative and human-centric innovation, because they are based on specific needs. In this way, AM is becoming a production technology tailored to specific social, environmental, and economic needs, which are therefore the basis of more sustainable development [57,58]. Although additive manufacturing (AM) has provided unprecedented opportunities in the development of human-centered products, i.e., customization, especially in medical applications, there are still many practical challenges

limiting broader adoption, especially in three main areas of industrial production, namely quality assurance, design optimization, and customization.

For example, the use of AI in design optimization for AM (DfAM) implies the optimization of the geometry of AM products to achieve the desired properties or minimize mass or cost. AI significantly accelerates design by, for example, integrating with support vector machine (SVM) methods to identify a subset of AM design features [59]. In DfAM, AI is also included in the materials design phase; for example, to be able to modify the materials used for traditional manufacturing techniques for AM processing [60]. The second fundamental aspect of HCAI in AM is the optimization in the choice of the AM processes and of the related process parameters. AI-based AM process optimization relies on multi-criteria approaches such as fuzzy logic operators [61] and hybrid schemes integrating multiple method selection. For optimizing AM parameters, there are several AI approaches that are based on the chosen variables, constraints, and objective functions. For example, there are many studies that simultaneously optimize the accuracy and mechanical properties of the product [62,63]. The third aspect directly influenced using AI is quality assessment. Fundamental, in fact, in the development of personalized human-centered products in AM, is the assessment, which can be made according to multiple aspects such as metrology, sustainability, dimensionality and ownership, widely explored in research [64–66].

Quality assurance is a fundamental step in industrial production; for example, in ensuring the performance of production lines, as well as for optimizing various materials and designs. AI-based systems can analyze data from IoT sensors within the AM process to identify anomalies or defects and quickly correct parameters or reprint defective parts. In design optimization, artificial intelligence can enhance AM in the design process, encouraging and further developing the design for AM (DfAM) and enabling significant progress in this regard [67]. AI algorithms can analyze large amounts of data to establish and adjust the factors that most affect design. The AM process can, therefore, easily execute enhanced designs that capture the exact specifications of the project. Additionally, artificial intelligence combined with AM enables manufacturers to create customized products, especially since AI algorithms can analyze customer data and preferences to create personalized designs, which can be easily implemented, and 3D printed by AM systems.

In addition to incentivizing an optimized product lifecycle management, AM technologies can definitely foster better resource efficiency, enhanced supply chains, and new business models, especially when combined with AI and the human-centered approach. Indeed, the supply chain can benefit from real-time decision-making, predictive analytics, and customized on-demand production towards greater sustainability and adaptability. In [68], key elements of human-centric supply chains are highlighted, placing people at the center of supply chain management. AI's ability to analyze vast amounts of data ensures more accurate forecasting and efficient logistics, while AM provides the flexibility to quickly respond to market changes and customer demands. Through the use of technology, the human-centric supply chain aims to create a sustainable society that prioritizes corporate social and environmental responsibility in supply chain development, while also addressing customer preferences and requirements throughout sourcing, production, and delivery of goods and services [13,69,70]. This capability not only offers new market opportunities but also supports innovative business models where manufacturers can offer premium quality assurance or customized services based on AI-driven insights. The aim of using AI in AM is to provide and manage intelligent products, services, and experiences while ensuring higher product standards and reducing waste. By providing enhanced product reliability and operational efficiency, the manufacturer not only attracts premium clients willing to pay for superior quality or customized products but also creates new revenue streams and competitive advantages in the market, and also revolutionizes trade and management practices towards more competitive and sustainable products and services [33,71].

Persistent challenges and forthcoming research prospects in these AM areas primarily include: the development of advanced AI methods oriented towards AM to handle data with high heterogeneity and dimensionality but low availability; the complete use of

physical knowledge of AI methods for personalized AM applications, while ensuring process safety and data privacy, and human-machine interaction aimed at integrating humans into the design, manufacturing, and quality control processes, through the use of interpretable AI, and a human-machine society empowered by IoT, AR/VR, etc.

A fundamental and prominent element for enabling intelligent and connected manufacturing through the implementation of sensors and AI is represented by the deployment of ideal 5G/6G networks for massive IoT implementations. These networks offer high-speed connectivity, low latency, and reliability, essential elements to ensure optimal integration of AI technologies in the context of data-driven additive manufacturing. They also ensure precise control of critical systems, data security, real-time automation, and adjustments, as well as efficient human-machine collaboration that requires low-latency communication. For example, AI-driven robots and machinery can work in collaboration with human workers, with 5G/6G ensuring that communication between these entities is nearly instantaneous. More importantly, 5G/6G networks are equipped with a powerful paradigm, which is edge intelligence, a form of distributed network intelligence built cooperatively through IoT data, wireless communications, edge computing capabilities, and artificial intelligence. Edge computing (in addition to classic cloud computing) in manufacturing environments is essential for (i) enabling synergy between users, personal devices, network infrastructure, and “things” such as cobots, machinery, etc.; (ii) providing computational capabilities to execute AI artifacts necessary for data-driven decisions and control processes.

The main challenges to address in this area, as summarized in Figure 3, include:

- **Network reliability and security:** Networks enabled by 5G/6G communication must be protected and highly configurable to meet AM requirements. For example, 5G and 6G networks extensively use network virtualization and programmability. This programmability comes with a cost, namely the centralization of network control. While there is the advantage that network and traffic control occur through a single entity, the disadvantage of this approach is the presence of a “single point of vulnerability” that can impact network availability, which is a critical concern in the AM network. If a controller malfunctions, it can affect the entire infrastructure, leading to a communication blackout. Moreover, in a virtualized infrastructure, there is a need to unify and synchronize security and access systems for both virtual network resources (links), cloud (servers, virtual machines, containers), and data storage (disks) to prevent malicious intrusions.
- **Integration of edge computing:** Artificial intelligence and AM require high data processing activity. Although this activity can be performed using cloud computing technologies, the bandwidth required to send data to the public or private cloud is not negligible. To overcome this issue, preprocessing nodes can be positioned at the edge of the network. Integrating these edge nodes and the impact they can have on network functionalities, such as latency and security, is an open challenge.
- **Deterministic latency:** 5G/6G networks and programmable networks can introduce some latency. For example, packets traversing within a network must be processed by an entity programmed to make traffic decisions. The path between switches and the programmed entity can be the source of unforeseen latencies, which can result in some synchronization losses in AM applications. On the other hand, efficient, secure, and real-time human-machine collaboration requires communication not only with low latency but also nearly deterministic latency, meaning a delay in data transfer that follows a predictable and constant pattern and can thus be predicted based on known factors or specific conditions within a system. Ensuring low and deterministic latency requires the implementation of specific network technologies and architectures and is still an open challenge in the industrial context.

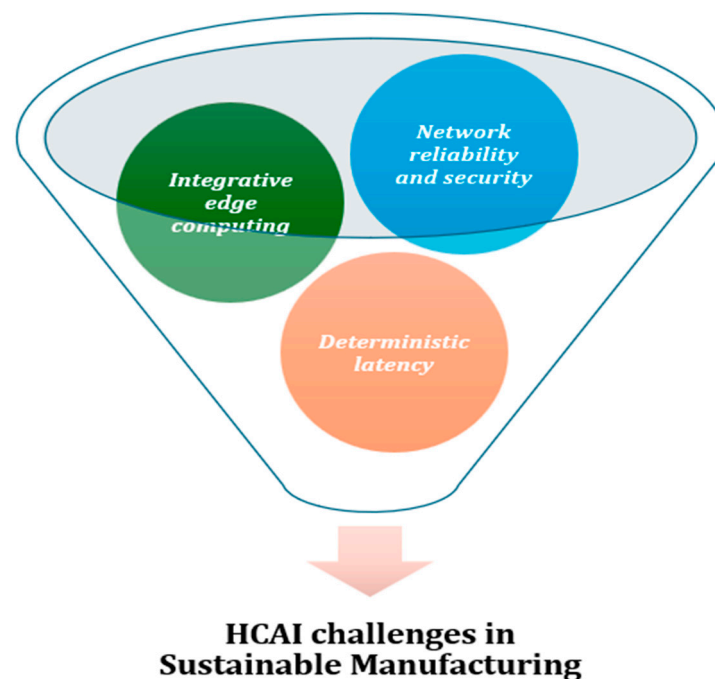


Figure 3. HCAI main challenges in sustainable manufacturing.

One possible approach to overcoming the challenges of AM discussed in this paper is to promote AM customization in the industrial sector by optimizing IoT sensors connected through high-performance, low-latency networks to improve the interaction between the customer and the finalization of the part, both from the design phase and during production. Specifically, at the 5G/6G network level, the proposed approach involves the use of virtualization technologies and synchronization protocols and low-latency networks, also through integration with industrial Ethernet networks to make the most of their respective characteristics. For example, to ensure security, a solution of distributed monitoring and recording of production data and machine operation status can be utilized. Additionally, employing a software virtualization technology, such as micro-containers or virtual machines, properly deployed in edge computing nodes, can ensure low-latency performance. Finally, there must be appropriate integration of industrial Ethernet and 5G networks, the former for fixed and wired applications with time-sensitive protocols that prioritize quick and timely packet scheduling, removing non-deterministic communication characteristics, and the latter for mobile applications and real-time wireless communications requiring precise response times, such as advanced machinery and robotics control.

Moreover, to enhance the integration of edge nodes with high-speed networks, an intent-based networking approach can be employed, which is an advanced approach for configuring and managing infrastructure based on defining user-desired goals and intentions, rather than manually configuring individual network devices. This can be enriched using artificial intelligence to address user profiling to enhance human-centered technical user-network interaction. Additionally, data security intents can also be designed to provide effective yet simplified support for network security. Similarly, blockchain technology can undoubtedly be useful in enhancing the cryptographic security and immutability of data as well as controlling data access. This is crucial for protecting sensitive data related to production and intellectual property from unauthorized access or fraudulent modifications that could compromise both computer and production systems, potentially leading to cybersecurity issues for both the systems and workers.

Different practical approaches or methods can be used to test and confirm, through real data and observations, the framework proposed. These include experiments, case studies, statistical analyses, surveys, just to name a few, that allow verifying whether the developed theories are actually reflected in reality [72,73]. As discussed in Section 2.2, various case

studies can be considered for validation across different manufacturing stages, while assessing the outcomes of applying HCAI in terms of process efficiency, human-machine collaboration, and sustainability.

4. Research Gap and Challenges for Industry 5.0 and Government 5.0

Beyond the technological challenges, another area of investigation should be devoted to political and social studies and challenges, particularly regarding improving safety and health in the workplace, especially in the manufacturing sector [74]. The development of an AI-based policy paradigm aligned with circular economy practices to adapt to Industrial Revolution 5.0 is critical, to steer the manufacturing sector in a sustainable, equitable, and unbiased direction.

Industry 5.0 needs Government 5.0 [75,76]. There is a significant discrepancy in the pace of change between businesses and the public sector. The uncertainty, instability, and rapid advancement of technologies require the public sector to respond with strategic agility. Just as businesses can achieve systemic transformation in short periods, so too must the public sector prepare adequate and timely decision-making tools. Given the complexity and scope of the revolution brought about by AI, it has become crucial to establish regulations and guidelines to govern its use. This involves issues such as ethics in AI, model transparency, legal responsibility, privacy protection, and security. In this regard, it may be necessary to establish standards and protocols for the evaluation and certification of AI to ensure compliance with ethical, regulatory, and security rules [77,78].

To increase the impact of the proposed solutions, efforts should be directed towards ensuring greater attention to sector regulation, which cannot be based on outdated and inflexible tools in the tradition of “command and control”, with a regulatory state imposing strict limits dictated by norms. New tools are associated with better quality regulation, such as adaptive and experimental regulation and regulatory sandboxing. In this sense, regulation should aim to improve how AI can be used to serve social objectives (e.g., reduce environmental impacts, create a healthy work environment) with the well-being of workers placed at the center of the production process. That is, AI is now also used to regulate and improve government oversight of businesses, focusing on their ability to prevent or mitigate the risk of worker incidents. The economic challenges in implementing HCAI can be referred to the initial investment costs which can be prohibitive for many companies and can determine a competitive gap between the bigger enterprises and the smaller ones. Moreover, there is a growing need for a workforce skilled in both AI technologies and the principles of human-centered design. Training such talent is both challenging and costly. Simultaneously, it is increasingly crucial to ensure that actions effectively address the real challenges brought about by the artificial intelligence complexity.

Decision-makers are required to address these challenges by adopting the following policy recommendations: (1) fostering innovation and technological advancement, investing in research and development and digital infrastructure, and encouraging public-private partnership; (2) relaunching economic policies for sustainable growth by supporting SMEs and investing in education and training programs to equip the workforce with skills needed for the digital economy; (3) adapting regulatory frameworks by adopting flexible regulatory approaches; (4) involving a diverse range of stakeholder in the decision-making process to ensure comprehensive and inclusive solutions.

5. Conclusions

The reference scenario for this study is therefore that of human-centered AI (HCAI) centered on the smart manufacturing area in line with the concept of Industry 5.0 and fostering the essence of the circular economy. Specifically, the focus is on the environment of data-driven additive manufacturing (AM), aiming to prioritize the well-being of workers and users through product customization, integration of humans in the production and quality control process, and improvement of regulations in this area.

Specifically, among the key technologies for achieving personalized and high-quality production, AM plays a fundamental role in modern industrial contexts, especially when combined with IoT and AI systems to pursue advanced intelligent production in line with the concept of HCAI towards Industry 5.0 scenarios. Together, IoT and AI technologies combined with AM can reshape traditional production into a more agile, human-centered, and data-driven process. Indeed, IoT can play a crucial role by serving as a data infrastructure that enhances AM processes, promoting superior quality, reducing material waste, and enabling rapid and safe production in AM processes. AM is a game-changer for sustainable production due to its inherent ability to produce almost clean shapes, with very high efficiency in the use of materials, as well as the customized and flexible production of complex objects on demand. AM processes integrate AI in different ways, primarily AI-enabled human-centric AM in terms of each individual phase, i.e., starting from design customization, optimization of different AM processes, and evaluation of quality. Moreover, intelligent systems can support humans to enhance the user experience and ensure worker well-being in the manufacturing sector. A key and significant element in enabling intelligent and connected production by IoT and AI systems is the contribution of 5G/6G networks: 5G/6G networks provide essential high-speed, low-latency connectivity and reliability for the seamless integration of AI technologies for data-driven AM, while ensuring precise control of critical systems, data security, real-time automation and regulation, and collaborative efficiency. There are many application scenarios for the presented approach; specific focuses have been identified that exemplify and are the subject of further investigation.

The first focus concerns applications in the logistics and supply chain sector, which has already been greatly impacted by the AM production approach, and where HCAI systems can completely change the approach and type of human work in such supply flows. A second focus is on the optimization and especially the customization of AM processes aimed at bioprinting, both in the biomedical and food sectors. At the same time, specific challenges need to be addressed, requiring advanced research activities in both the AM field and the enabling technologies of 5G/6G networks and edge computing. Furthermore, these challenges are reflected in the corresponding challenges of the 5G/6G network to ensure the high availability and low-latency data transmission and massive data computation required by advanced automation and artificial intelligence algorithms.

The latter require computational capabilities to support automated control processes based on the timely processing of large amounts of data. Therefore, solutions need to be proposed for network reliability and security, minimal and deterministic latency, network, and edge computing integration.

In the context of the above challenges, the deployment of private 5G networks can offer interesting opportunities by providing advanced solutions to address challenges and improve operational efficiency. Private 5G networks enable dedicated radio coverage, targeted and guaranteed use of network resources aimed at the offered radio service, and the ability to use customer cloud infrastructures to realize an effective edge architecture within a specific production installation or manufacturing plant. These private 5G networks are independent of public networks and are created to meet the specific needs and challenges of advanced manufacturing applications, which represent the most widespread area of interest.

Looking at Government 5.0, policymaking is a highly intricate process that takes place in a dynamic environment, influencing the three pillars of sustainable development: society, economy, and environment. Decision-making processes encompass complex scenarios, which take place in rapidly changing and uncertain environments, and involve conflicts among various interests [79]. Every political decision triggers social reactions, impacts economic and financial factors, and has significant environmental consequences. Enhancing decision-making in this context can lead to substantial benefits across all these areas. In this sense, the role of public institutions, and their ability to respond to challenges arising from innovation, represents a focal point to ensure that private actors have an adaptive and non-burdensome regulatory environment capable of meeting the needs dictated by AI

development, as well as the definition of standards and protocols for the evaluation and certification of AI to ensure compliance with ethical, regulatory, and safety rules.

Finally, promoting effective communication and cooperation between different fields to advance human-centered artificial intelligence (HCAI) involves several strategies and practices. Some key approaches can be highlighted: cross-disciplinary research projects should include representatives from diverse disciplines, shared platforms and tools must facilitate collaboration and communication, and policy and governance initiatives should ensure that policymaking bodies and regulatory committees include representatives from various disciplines to create balanced and comprehensive regulations for HCAI. By implementing these strategies, communication and cooperation between different fields can be significantly enhanced, leading to more comprehensive and effective advancements in human-centered artificial intelligence.

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